Egg Drop Strategies for Four Eggs: Corrected Asymptotic Constants for the 1-3 and 2-2 Approaches

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Abstract

A previous analysis of the 1-3 and 2-2 strategies for four eggs used the non-tight bound $P_1(k) \le kN^{1/k}$ and assumed uniform intervals for the first egg. That is incorrect. The 1-3 strategy means that the first egg is used to bracket the critical floor into **optimally chosen intervals**, after which the remaining three eggs solve the reduced problem. Using the exact minimax capacity

$$E(d,k) = \sum_{j=1}^{k} \binom{d}{j},$$

we show that the 1-3 strategy actually achieves the full four-egg capacity. Its asymptotic constant is

$$24^{1/4} \approx 2.213.$$

The 2-2 strategy, which uses two eggs on superfloors and two eggs inside a block, has asymptotic constant

$$(64)^{1/4} = 2\sqrt{2} \approx 2.828.$$

Thus the **1-3 strategy is asymptotically better** for four eggs and a large number of floors.

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1. Introduction

The classical egg drop problem asks for the minimum number of drops needed to identify the critical floor in an N-story building with k eggs. The paper *Egg Drop Problems: They Are All They Are Cracked Up to Be!* proved the convenient upper bound

$$P_1(k) \le \lceil kN^{1/k} \rceil.$$

This bound is **not tight**. The exact minimax capacity is given by the binomial sum

$$E(d,k) = \sum_{j=1}^{k} \binom{d}{j},$$

the maximum number of floors testable with d drops and k eggs. For large N,

$$E(d,k) \sim \frac{d^k}{k!},$$

so the true asymptotic constant is $(k!)^{1/k}$, not k. For $k = 4$, the correct constant is

$$(24)^{1/4} \approx 2.213,$$

whereas the earlier bound gave 4.

In this note we re-analyze the two strategies for four eggs using the exact capacity. In particular, we correct the treatment of the 1-3 strategy: the first egg must be dropped at the **optimal cumulative intervals** determined by the remaining three-egg capacity, not at uniform jumps.

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2. Exact Minimax Capacity

Let $E(d,k)$ be the maximum number of floors that can be tested with d drops and k eggs. The standard recurrence is

$$E(d,k) = E(d-1,k-1) + 1 + E(d-1,k),$$

with $E(0,k) = 0$ and $E(d,0) = 0$. Its solution is

$$E(d,k) = \sum_{j=1}^{k} \binom{d}{j}.$$

For large d,

$$E(d,k) = \binom{d}{k} + O(d^{k-1}) \sim \frac{d^k}{k!}.$$

Hence the minimum number of drops to cover N floors with k eggs is the smallest d such that

$$\sum_{j=1}^{k} \binom{d}{j} \geq N.$$

For $k = 4$, this becomes

$$E(d,4) = \binom{d}{1} + \binom{d}{2} + \binom{d}{3} + \binom{d}{4} = d + \frac{d(d-1)}{2} + \frac{d(d-1)(d-2)}{6} +$$

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3. The 1-3 Strategy (Optimal Intervals)

Strategy. Use one egg to bracket the critical floor. Suppose we have D total drops. If the first egg breaks on the t-th drop, we have used t drops and have $D - t$ drops left with three eggs. The interval between the $(t-1)$-th and t-th drop must be resolvable by three eggs with $D - t$ drops.

With r drops and three eggs, the maximum number of floors that can be tested is

$$E(r,3) = \binom{r}{1} + \binom{r}{2} + \binom{r}{3}.$$

The number of possible critical floors in an interval of size x is x. To resolve x possibilities with r drops and three eggs, we need

$$E(r,3) + 1 \geq x,$$

because $E(r,3)$ is the maximum number of floors N such that $N+1$ outcomes can be distinguished. Thus the maximum interval size that can be handled with r drops is

$$E(r,3) + 1.$$

If the first egg breaks on the first drop, the interval below must have size at most $E(D-1,3)+1$. If it breaks on the second drop, the next interval must have size at most $E(D-2,3)+1$, and so on. If it survives all D drops, the last interval above must be empty, so the total number of floors covered is

$$\sum_{r=0}^{D-1} \big(E(r,3) + 1\big).$$

Now compute:

$$\sum_{r=0}^{D-1} E(r,3) = \sum_{r=0}^{D-1} \left[\binom{r}{1} + \binom{r}{2} + \binom{r}{3}\right] = \binom{D}{2} + \binom{D}{3} + \binom{D}{4}.$$

Adding the D ones gives

$$\sum_{r=0}^{D-1} \big(E(r,3) + 1\big) = \binom{D}{1} + \binom{D}{2} + \binom{D}{3} + \binom{D}{4} = E(D,4).$$

Therefore the 1-3 strategy with optimally chosen intervals **achieves the full four-egg capacity**:

$$\text{Capacity}_{\text{1-3}}(D) = E(D, 4).$$

The minimum number of drops for N floors is the smallest D with

$$E(D, 4) \geq N.$$

Asymptotically,

$$E(D, 4) \sim \frac{D^4}{24},$$

so

$$D \sim (24N)^{1/4} = 24^{1/4} N^{1/4} \approx 2.213\, N^{1/4}.$$

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4. The 2-2 Strategy

Strategy. Partition the N floors into M blocks of size F. Use two eggs to find the critical superfloor among the M blocks, then use two eggs to find the exact floor inside the critical block of size F.

Suppose the total number of drops is D. Allocate d_1 drops to the first phase and d_2 drops to the second phase, with $d_1 + d_2 = D$. With two eggs, the maximum number of floors testable with d drops is

$$E(d, 2) = \binom{d}{1} + \binom{d}{2} = d + \frac{d(d-1)}{2} = \frac{d(d+1)}{2}.$$

Thus the first phase can handle at most

$$M = E(d_1, 2)$$

superfloors, and the second phase can handle at most

$$F = E(d_2, 2)$$

floors per block. The total number of floors covered is

$$N = M \cdot F = E(d_1, 2)\, E(d_2, 2).$$

For large d,

$$E(d, 2) \sim \frac{d^2}{2}.$$

Hence

$$N \sim \frac{d_1^2}{2} \cdot \frac{d_2^2}{2} = \frac{d_1^2 d_2^2}{4}.$$

Maximize $d_1^2 d_2^2$ subject to $d_1 + d_2 = D$. By symmetry or calculus, the maximum occurs at

$$d_1 = d_2 = \frac{D}{2}.$$

Then

$$N \sim \frac{(D/2)^4}{4} = \frac{D^4}{64}.$$

Therefore

$$D \sim (64N)^{1/4} = 64^{1/4} N^{1/4} = 2\sqrt{2}\, N^{1/4} \approx 2.828\, N^{1/4}.$$

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5. Comparison

Strategy	Asymptotic drops
1-3 (optimal intervals)	$24^{1/4} N^{1/4} \approx 2.213\, N^{1/4}$
2-2	$64^{1/4} N^{1/4} = 2\sqrt{2}\, N^{1/4} \approx 2.828\, N^{1/4}$

The ratio is

$$\frac{2.828}{2.213} \approx 1.278.$$

Thus the 1-3 strategy uses about **22% fewer drops** asymptotically than the 2-2 strategy.

This corrects the earlier conclusion that both strategies were asymptotically equivalent with constant 4. It also corrects the intermediate value 2.746 that arose from assuming uniform intervals for the first egg. The correct constant for 1-3 is $24^{1/4} \approx 2.213$.

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6. Exact Integer Optimization and Examples

For a given N, the exact minimum drops are:

- **1-3:** smallest D such that $E(D, 4) \geq N$.
- **2-2:** smallest D such that $\max_{d_1 + d_2 = D} E(d_1, 2) E(d_2, 2) \geq N$.

Some values:

N	1-3 drops	2-2 drops
100	8	8
1000	13	15
10^6	70	90

For $N = 1000$, the 1-3 strategy needs 13 drops while the 2-2 strategy needs 15. For $N = 10^6$, the gap is 70 versus 90.

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7. Conclusion

The 1-3 strategy for four eggs, when the first egg is dropped at **optimally chosen intervals**, is exactly the optimal four-egg strategy. Its asymptotic constant is $24^{1/4} \approx 2.213$. The 2-2 strategy is suboptimal, with constant $2\sqrt{2} \approx 2.828$. The earlier bound $P_1(k) \leq kN^{1/k}$ is not tight and should not be used for constant-factor comparisons.

References

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